



Design project proposal

Comparison of flow prediction models
for the Arve River in Geneva



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1 Introduction

Flood forecasting for the Arve River in Geneva is a critical component of regional risk management. The centennial flood of May 2015, followed by the major event of November 2023, clearly illustrated how extreme discharges can affect the safety of the population, urban infrastructure and key services, while also showing the value of timely and accurate forecasts to support emergency response and reservoir management. At the same time, these events revealed limitations in existing forecasting tools, particularly regarding the anticipation of flood peaks, their timing, and the explicit treatment of forecast uncertainty.

In response to these challenges, several hydrometeorological forecasting systems have been developed or upgraded for the Arve at Geneva by different institutions over the last decade, including federal, cantonal and local actors as well as partners in neighbouring France. These systems are based on diverse modelling approaches, ranging from physically based rainfall-runoff models to machine-learning methods, and they rely on different combinations of meteorological inputs, spatial discretisations and post-processing strategies. As a result, they can produce forecasts with different characteristics and levels of confidence for the same hydrological situation.

However, despite the growing number and sophistication of these tools, there is currently no consistent, transparent framework that allows their performance to be compared over recent years or that clarifies how their differences should be interpreted by operational users during flood situations. This raises important questions for the CEL-SUCR, the agency in charge of flood risk management of the Arve in Geneva: how much trust to place in each system, how to interpret diverging forecasts, and how to make best use of the available information when taking time-critical decisions.

The Arve River basin is a high-relief Alpine catchment that stretches from the Mont-Blanc massif to the city of Geneva, where it joins the Rhône. The river rises on the French side of the Col de Balme, in the Mont-Blanc massif, and flows 108 km before reaching the Rhône at La Jonction in Geneva, at an elevation of about 370 m. The basin extends over just above 2 000 km², with minimum elevations around 370 m and maximum elevations above 4 800 m (Mont-Blanc), and an average basin elevation of about 1 370 m, so that more than half of the basin area lies above 1 000 m and about 5% is glacierised. This strong elevation gradient produces a predominantly nival to glacio-nival regime in the upper reaches, gradually evolving downstream into a more mixed pluvio-nival behaviour, and leads to marked seasonal variations in discharge with high flows typically in late spring–summer and lower flows in winter [14].

For the purposes of this project, discharge data and operational forecasts are available at the Swiss hydrometric station “Arve, Genève Bout du Monde”, located just upstream of the confluence with the Rhône. At this station, the contributing drainage area is about 1 970–1 980 km², and it is the main reference point used by the Canton of Geneva and federal authorities for monitoring floods and managing flood risk along the Arve and in central Geneva. This location integrates the contributions of the major Alpine tributaries (including the Giffre and Borne on the French side) and is directly relevant for assessing the impact of floods on the urban area and on the downstream Rhône system [4].

2 Literature Review

Hydrological models for rainfall-runoff simulation are generally classified into three broad categories based on their spatial representation: lumped (or “global”) models, which treat the entire catchment as a single unit with spatially averaged inputs and parameters; semi-distributed models, which partition the basin into sub-basins or elevation bands and apply lumped process descriptions within each unit; and fully distributed models, which explicitly resolve spatial variability of inputs, states, and parameters on a grid or mesh [1]. Semi-distributed, elevation-band models are particularly well suited to Alpine basins with strong snow and glacier influence, while fully distributed models are often used when detailed spatial patterns of precipitation, land cover, or topography need to be represented. In the context of Alpine catchments like the Arve, comparisons of models with varying complexity have shown that semi-distributed approaches can effectively capture snowmelt and glacier dynamics, though performance depends on calibration and input resolution [2].

Traditional model intercomparison studies provide valuable benchmarks for evaluating flood simulation capabilities. For instance, Li and Zhang (2008) compared three GIS-based distributed hydrological models (a grid-and-

topography-based model, a GIS-enhanced TOPMODEL, and a distributed Xinanjiang model) in a mountainous sub-basin of the Yellow River, using metrics such as relative peak flow error, relative runoff error, peak timing error, and Nash-Sutcliffe efficiency [9]. The study highlighted the advantages of distributed representations for capturing spatial heterogeneity in runoff generation, with the grid-based model performing best overall. Similarly, Zhai et al. (2021) evaluated three distributed models (TOPMODEL, HEC-HMS, and CNFF) across 13 humid and semi-humid mountainous catchments in China, focusing on flash flood reproduction with metrics including relative runoff error (RER), relative peak error (REQ), peak timing error (TP), and scaled Nash-Sutcliffe efficiency (SNSE) [15]. They found that model performance was modulated by catchment attributes such as drainage area, slope, forest cover, and rain gauge density, with hybrid approaches showing promise for operational use.

Recent advances have increasingly integrated machine learning (ML) with physical models to improve flood forecasting, particularly in data-rich operational settings. Mahmoodi et al. (2024) compared HEC-HMS (semi-distributed hydrological model), TOPSIS (multi-criteria decision-making based on morphometrics), and artificial neural networks (ANN) for identifying flood source areas in an Iranian catchment, finding that ANN closely reproduced HEC-HMS classifications (correlation ≈ 0.99) while being more robust in data-scarce conditions [10]. Hybrid ML frameworks are particularly relevant for systems like those used for the Arve, where physics-based pre-processing feeds into neural networks: Wang et al. (2024) proposed a grid-based runoff generation model coupled with LSTM for flood forecasting, demonstrating superior skill during extreme events [13], while Vionnet et al. (2023) applied LSTM to short-term runoff prediction in a Swiss Alpine catchment, outperforming traditional models like MINERVE during snowmelt-driven floods [12].

Another key distinction is to be made between deterministic and probabilistic forecasting. Deterministic models produce a single discharge trajectory, offering an interpretable best-guess estimate but no explicit uncertainty propagation. Some systems complement this with a confidence interval derived from historical forecast errors at each horizon, though such intervals reflect average past performance rather than the specific meteorological uncertainty of a given forecast situation. Probabilistic systems instead propagate an ensemble of meteorological scenarios through the hydrological model, yielding a distribution of plausible discharge outcomes from which exceedance probabilities can be derived [3]. Verification of probabilistic forecasts remains essential, the ECMWF guide (2023) details the use of Continuous Ranked Probability Score (CRPS) and related metrics for assessing ensemble reliability in hydrological applications [5].

3 Goals

The primary objective of this semester project is to propose a methodology for evaluating the quality of operational discharge forecasts for the Arve River at Geneva and to apply it systematically to existing forecasting systems over the 2020–2024 period. Three specific goals support this objective:

1. Model inventory and comparison : Document the structure, input data requirements, forecasting horizons, and output types (deterministic vs. probabilistic) of the main operational systems, identifying commonalities and structural differences with potential implications for forecast skill and operational usability.
2. Performance evaluation and ranking : Select and implement appropriate verification metrics for both deterministic (e.g., Nash-Sutcliffe Efficiency, timing errors) and probabilistic forecasts (e.g., CRPS, Brier Skill Score), compute performance scores across the full period with emphasis on high-flow events, and rank systems according to their strengths for different lead times and flood situations.
3. Operational recommendations : Synthesize evaluation results to provide practical guidance for the Geneva Cantonal Water Office (OCEau), including preferred systems by use case, strategies for combining forecasts from multiple sources, and approaches to handle and communicate forecast uncertainty in real-time decision-making.

4 Models considered

The four forecasting systems evaluated in this project are:

- The OFEV system, using the WaSiM physics-based distributed model which is forced with multiple meteorological inputs including deterministic (COSMO-1/ICON-1) and ensemble (COSMO-E/ICON-2) forecasts from MeteoSwiss, as well as ECMWF Integrated Forecast System [11].

- The Hydrique Routing System (RS) model, a semi-distributed conceptual model operated by Hydrique Ingénieurs. Based on functional modelling to reproduce watershed processes, it integrates various water regulation structures and is built on the GSM-SOCONT concept [7].
- The Hydrique Machine Learning model, a data-driven MLP/LSTM system also operated by Hydrique Ingénieurs, using the RS physical model state variables as additional inputs [8].
- The CNR (Compagnie Nationale du Rhône) forecasting chain, which includes a deterministic hourly tool alongside a probabilistic ensemble-based chain [6]. The current operational status of the probabilistic component and its specific application to the Arve at Geneva are pending further documentation from the partner.

5 Calendar

The following table provides an indicative overview of the project timeline. Given the exploratory nature of the work tasks may be adjusted as the project progresses.

Weeks (EPFL calendar)	Dates	Tasks and deadlines
1-3	16.02 - 08.03	First meeting with professor and external partner Literature review and getting to know the forecasting models Project proposal writing
4-5	09.03 - 22.03	In-depth review of the forecasting models and their characteristics Definition of evaluation metrics and performance indicators Meeting with CNR representative
6-10	23.03 - 03.05	W6-7: Meeting with Hydrique and BAFU representatives W8: Intermediate report submission Implementation of evaluation metrics and models assessment
11-13	04.05 - 24.05	Models comparison and ranking Synthesis of results and operational recommendations W13: Report submission for comments before finalization
14-16	25.05 - 12.06	W14: Poster presentation W15: Oral presentation and final report submission

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